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年級：

科目：

## 練習題 17.2

$$2. (1) \int_{\Gamma} \frac{x dy - y dx}{x^2 + y^2} = \frac{1}{a} \int_{\Gamma} \left( -\frac{y}{\sqrt{x^2 + y^2}}, \frac{x}{\sqrt{x^2 + y^2}} \right) \cdot (dx, dy) = \frac{1}{a} \int_{\Gamma} \left( -\frac{y}{\sqrt{x^2 + y^2}}, \frac{x}{\sqrt{x^2 + y^2}} \right) \cdot \frac{1}{a} (-y, x) ds$$

$$= \frac{1}{a^2} \int_{\Gamma} \sqrt{x^2 + y^2} ds = \frac{1}{a} \int_{\Gamma} ds = \frac{2\pi a}{a} = 2\pi.$$

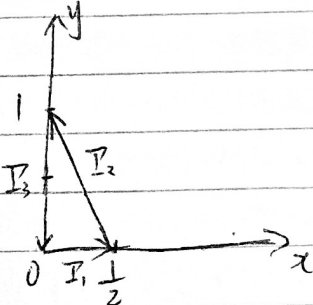
$$(2) \int_{\Gamma} (x+y) dx + (x-y) dy = \int_0^{2\pi} [(a \cos t + b \sin t) a (-\sin t) + (a \cos t - b \sin t) \cdot b \cos t] dt$$

$$= \int_0^{2\pi} \left[ \frac{(-a^2 - b^2)}{2} \sin 2t + ab \cos 2t \right] dt = 0$$

$$(3) \int_{\Gamma} (x^2 - 2xy) dx + (y^2 - 2xy) dy = \int_{-1}^1 [(y^4 - 2y^3) \cdot 2y + (y^2 - 2y^3)] dy = \int_{-1}^1 (2y^5 - 4y^4 - 2y^3 + y^2) dy$$

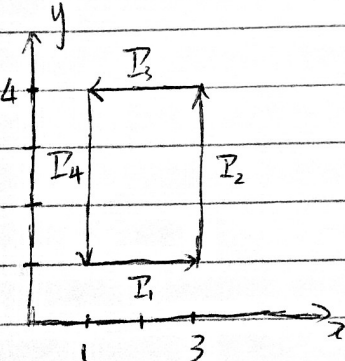
$$= 2 \int_0^1 (y^2 - 2y^4) dy = 2 \times \left( \frac{1}{3} - \frac{4}{5} \right) = -\frac{14}{15}$$

(4)



$$\int_{\Gamma} x dy = \int_{\Gamma_1} + \int_{\Gamma_2} + \int_{\Gamma_3} = 0 + \int_{\frac{1}{2}}^0 x \cdot (-2 dx) + 0 = x^2 \Big|_0^{\frac{1}{2}} = \frac{1}{4}$$

(5)



$$\int_{\Gamma} (x^2 + y^2) dy = \int_{\Gamma_1} + \int_{\Gamma_2} + \int_{\Gamma_3} + \int_{\Gamma_4} = \int_{\Gamma_2} (x^2 + y^2) dy + \int_{\Gamma_4} (x^2 + y^2) dy$$

$$= \int_1^4 (3^2 + y^2) dy + \int_4^1 (1^2 + y^2) dy = \int_1^4 8 dy = 24$$

$$3. \int \frac{x dy - y dx}{a x^2 + 2 b x y + c y^2} = \int_0^{2\pi} \frac{\cos^2 \theta + \sin^2 \theta}{a \cos^2 \theta + b \sin 2\theta + c \sin^2 \theta} d\theta = \int_0^{2\pi} \frac{1}{\frac{a}{2}(1+\cos 2\theta) + b \sin 2\theta + \frac{c}{2}(1-\cos 2\theta)} d\theta$$

$$= 2 \int_0^{2\pi} \frac{1}{(a-c) \cos 2\theta + 2b \sin 2\theta + (a+c)} d\theta = 2 \int_0^{2\pi} \frac{1}{(a-c) \cos \theta + 2b \sin \theta + (a+c)} d\theta = \textcircled{1}$$

當  $a > c$  時,  $\textcircled{1} = 2 \int_0^{2\pi} \frac{d\theta}{\sqrt{(a-c)^2 + 4b^2} \cos(\theta - \arctan(\frac{2b}{a-c})) + (a+c)}$

$$= 2 \int_0^{2\pi} \frac{1}{\sqrt{(a-c)^2 + 4b^2} \cos \theta + (a+c)} d\theta$$

設  $t = \tan \frac{\theta}{2} \Rightarrow dt = \frac{1}{2}(t^2+1) d\theta \Leftrightarrow d\theta = \frac{2dt}{t^2+1}$

$$\textcircled{1} = 4 \left[ \int_0^{+\infty} \frac{1}{\sqrt{(a-c)^2 + 4b^2} \cdot \frac{1-t^2}{1+t^2} + (a+c)} \cdot \frac{dt}{t^2+1} + \int_{-\infty}^0 \frac{1}{\sqrt{(a-c)^2 + 4b^2} \cdot \frac{1-t^2}{1+t^2} + (a+c)} \cdot \frac{dt}{t^2+1} \right]$$

$$= 8 \int_0^{+\infty} \frac{dt}{[(a+c) - \sqrt{(a-c)^2 + 4b^2}] t^2 + [(a+c) + \sqrt{(a-c)^2 + 4b^2}]}$$

$$= \frac{8}{(a+c) + \sqrt{(a-c)^2 + 4b^2}} \sqrt{\frac{(a+c) + \sqrt{(a-c)^2 + 4b^2}}{(a+c) - \sqrt{(a-c)^2 + 4b^2}}} \cdot \frac{\pi}{2} = \frac{4\pi}{(a+c) + \sqrt{(a-c)^2 + 4b^2}} \sqrt{\frac{(a+c) + \sqrt{(a-c)^2 + 4b^2}}{(a+c) - \sqrt{(a-c)^2 + 4b^2}}}$$

$$= 4\pi \operatorname{sgn}(a+c) \sqrt{\frac{1}{4ac-4b^2}} = 2\pi \operatorname{sgn}(a^2-c^2) \sqrt{\frac{1}{ac-b^2}}$$

當  $a < c$  時, 則  $-a > -c$ ,  $\textcircled{1} = -2 \int_0^{2\pi} \frac{d\theta}{[(c-a) - (-c)] \cos \theta + 2(-b) \sin \theta + [(c-a) + (-c)]}$

$$= 2\pi \operatorname{sgn}(-a-c) \sqrt{\frac{1}{ac-b^2}} = -2\pi \operatorname{sgn}(a+c) \sqrt{\frac{1}{ac-b^2}} = 2\pi \operatorname{sgn}(a^2-c^2) \sqrt{\frac{1}{ac-b^2}}$$

當  $a = c$  時, 定義此時  $\arctan(\frac{2b}{a-c}) = \frac{\pi}{2}$ , 便有  $\textcircled{1} = 2\pi \operatorname{sgn}(0) \sqrt{\frac{1}{ac-b^2}} \quad (\operatorname{sgn}(0) = 1)$

所以原積分  $= 2\pi \operatorname{sgn}(a^2-c^2) \sqrt{\frac{1}{ac-b^2}}$

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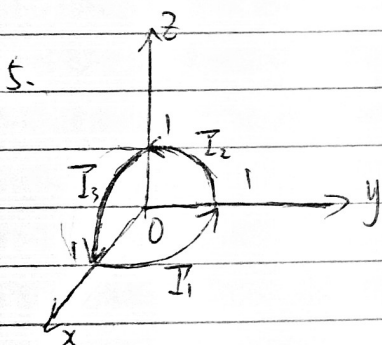
$$4.(1) \int_I xz^2 dx + yx^2 dy + zy^2 dz = \int_0^1 (t^7, t^4, t^7) \cdot (1, 2t, 3t^2) dt = \int_0^1 (3t^9 + t^7 + 2t^5) dt$$

$$= \frac{3}{10} + \frac{1}{8} + \frac{1}{3} = \frac{36+15+40}{120} = \frac{91}{120}$$

$$(2) \int_I (y+z) dx + (z+x) dy + (x+y) dz = \int_0^\pi (a \sin 2t + a \cos^2 t, a, a \sin^2 t + a \sin 2t) \cdot (a \sin 2t, 2a \cos 2t, -a \sin 2t) dt$$

$$= a^2 \int_0^\pi (\sin^2 2t + \sin 2t \cos^2 t + 2 \cos 2t - \sin 2t \sin^2 t - \sin^2 2t) dt$$

$$= a^2 \int_0^\pi (\frac{1}{2} \sin 4t + 2 \cos 2t) dt = 0$$



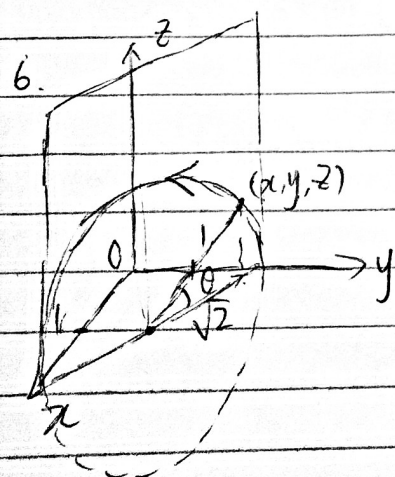
$$\int_I = \int_{I_1} y^2 dx - x^2 dy + \int_{I_2} z^2 dy - y^2 dz + \int_{I_3} x^2 dz - z^2 dx$$

$$= 3 \int_{I_1} y^2 dx - x^2 dy = 3 \int_0^{\frac{\pi}{2}} (-\sin^3 t - \cos^3 t) dt$$

$$= 3 \left[ \int_0^{\frac{\pi}{2}} (1 - \cos^2 t) d(\cos t) - \int_0^{\frac{\pi}{2}} (1 - \sin^2 t) d(\sin t) \right]$$

$$= 3 \left[ \int_1^0 (1 - x^2) dx - \int_0^1 (1 - x^2) dx \right] = -6 \int_0^1 (1 - x^2) dx$$

$$= -6 \times \left( 1 - \frac{1}{3} x^3 \Big|_0^1 \right) = -6 \times \frac{2}{3} = -4$$



$$x = \sqrt{2} \cos \theta \sin 45^\circ + 1 = 1 - \cos \theta, \quad y = \sqrt{2} \cos \theta \cos 45^\circ + 1 = 1 + \cos \theta$$

$$z = \sqrt{2} \sin \theta$$

$$\int_I y dz + z dy + x dz = \int_0^{2\pi} [(1 + \cos \theta) \sin \theta - \sqrt{2} \sin^2 \theta + (1 - \cos \theta) \cdot \sqrt{2} \cos \theta] d\theta$$

$$= \int_0^{2\pi} -\sqrt{2} d\theta = -2\sqrt{2}\pi$$

7.  $x = \cos t, y = \sin t, z = \frac{1}{2} \sin 2t$

$$\int_{\Gamma} y dx + z dy + x dz = \int_0^{2\pi} (-\sin^2 t + \frac{1}{2} \sin 2t \cos t + \cos t \cos 2t) dt = \int_0^{2\pi} \frac{\cos 2t - 1}{2} dt$$

$$= -\pi$$

8.  $z^2 - a^2 = ax \Leftrightarrow x = \frac{1}{a}(a^2 - z^2)$

設  $z = a \sin \theta, x = a \cos^2 \theta$

$$y^2 = a^2 - z^2 - x^2 = a^2 \cos^2 \theta - a^2 \cos^4 \theta = a^2 \cos^2 \theta (1 - \cos^2 \theta) = \frac{a^2}{4} \sin^2 2\theta \Rightarrow y = \frac{a}{2} \sin 2\theta$$

$$\int_{\Gamma} \vec{F} \cdot d\vec{p} = \int_{\pi}^0 (\frac{a^2}{4} \sin^2 2\theta, a^2 \sin^2 \theta, a^2 \cos^4 \theta) \cdot (-2a \sin 2\theta, a \cos 2\theta, a \cos \theta) d\theta$$

$$= \int_{\pi}^0 (-\frac{a^3}{2} \sin^3 2\theta + a^3 \sin^2 \theta \cos 2\theta + a^3 \cos^5 \theta) d\theta$$

$$= a^3 \int_0^{\pi} (\frac{1}{2} \sin^3 2\theta - \frac{1 - \cos 2\theta}{2} \cos 2\theta + \cos^5 \theta) d\theta$$

$$= \frac{a^3}{2} \int_0^{\pi} \cos^2 2\theta d\theta = \frac{a^3}{2} \int_0^{\pi} \frac{1 + \cos 4\theta}{2} d\theta = \frac{a^3}{4} \pi$$

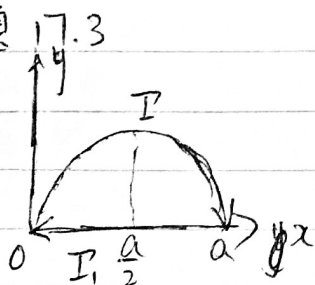
9. ~~設加速度是~~

$$\vec{F} = m \frac{d\vec{v}}{dt} \Rightarrow \int_{\Gamma} \vec{F} \cdot d\vec{p} = \int_{\Gamma} m \frac{d\vec{v}}{dt} \cdot d\vec{p}$$

$$\text{又知 } \vec{v} = \frac{d\vec{p}}{dt}, \text{ 所以 } \int_{\Gamma} m \frac{d\vec{v}}{dt} \cdot \vec{v} = m \int_{\Gamma} \vec{v} \cdot d\vec{v} = \frac{1}{2} m (\vec{v} \cdot \vec{v}) \Big|_{\vec{a}}$$

### 練習題

1. (3)



$$\partial D = -I_1 - I_2$$

$$\Rightarrow \int_{\Gamma} = \int_{\partial D} = \int_{-I_1} - \int_{I_2} = - \int_{\partial D} (e^x \cos y - e^x \cos y) dx dy + \int_0^a (0 dx + 0)$$

$$= 0$$



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$$2. (2) \text{ 令 } y = x \tan \theta \Rightarrow \frac{x^4}{\cos^4 \theta} = a^2 x^2 (1 - \tan^2 \theta) \Leftrightarrow x^4 = a^2 x^2 \cos^2 \theta (\cos^2 \theta - \sin^2 \theta)$$

$$\Leftrightarrow x^2 = a^2 \cos^2 \theta \cos 2\theta \Rightarrow x = a \cos \theta \sqrt{\cos 2\theta}$$

$$\Leftrightarrow y^2 = a^2 \sin^2 \theta \cos 2\theta \Rightarrow y = a \sin \theta \sqrt{\cos 2\theta}$$

$$2\theta \in [2k\pi - \frac{\pi}{2}, 2k\pi + \frac{\pi}{2}] \Leftrightarrow \theta \in [k\pi - \frac{\pi}{4}, k\pi + \frac{\pi}{4}]$$

$$\begin{aligned} G(D) &= \int_{\partial D} x dy = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} a^2 \cos \theta \sqrt{\cos 2\theta} \left( \cos \theta \sqrt{\cos 2\theta} + \sin \theta \cdot \frac{-2 \sin \theta}{2 \sqrt{\cos 2\theta}} \right) d\theta + \int_{\frac{3\pi}{4}}^{\frac{5\pi}{4}} \dots \\ &= 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} a^2 \left[ \cos^2 \theta \cos 2\theta - \frac{1}{2} \sin^2 2\theta \right] d\theta \\ &= 2a^2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left( \frac{1}{2} (\cos 2\theta - \sin^2 2\theta) + \frac{1}{2} \cos 2\theta \right) d\theta \\ &= a^2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\cos 4\theta + \cos 2\theta) d\theta = \frac{a^2}{2} \sin 2\theta \Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}} = a^2 \end{aligned}$$

(4) 記  $\frac{x}{a} = x_1, \frac{y}{b} = y_1$ .

$$x_1^{2n+1} + y_1^{2n+1} = c x_1^n y_1^n, \text{ 設 } y_1 = t x_1$$

$$(1+t^{2n+1}) x_1^{2n+1} = c t^n x_1^{2n} \Leftrightarrow \frac{1}{x_1} = \frac{1+t^{2n+1}}{c t^n} \Leftrightarrow x_1 = \frac{c t^n}{1+t^{2n+1}}, y_1 = \frac{c t^{n+1}}{1+t^{2n+1}}$$

$$\begin{aligned} G(D) &= \int_{\partial D} x dy = \int_0^{\infty} \frac{a c t^n}{1+t^{2n+1}} \cdot b c \frac{(n+1) t^n (1+t^{2n+1}) - t^{n+1} \cdot (2n+1) \cdot t^{2n}}{(1+t^{2n+1})^2} dt \\ &= a b c^2 \int_0^{\infty} \frac{t^{2n} (n+1 + n t^{2n+1} - 2n t^{2n+1} - t^{2n+1})}{(1+t^{2n+1})^2} dt \\ &= a b c^2 \int_0^{\infty} \frac{t^{2n} (n+1) - n t^{4n+1}}{(1+t^{2n+1})^2} dt \\ &= a b c^2 \left[ \frac{n+1}{2n+1} \int_0^{\infty} \frac{1}{(1+x)^2} dx - \frac{n}{4n+2} \int_0^{\infty} \frac{1}{(1+x)^2} dx \right] \end{aligned}$$

$$= abc^2 \left[ \frac{n+1}{4n+2} - \frac{2n}{4n+2} \int_0^{+\infty} \frac{x}{(1+x)^3} dx \right] = abc^2 \left[ \frac{n+1}{4n+2} + \frac{n}{4n+2} \left( \frac{x}{(1+x)^2} \right) \Big|_0^{+\infty} - \int_0^{+\infty} \frac{1}{(1+x)^2} dx \right]$$

$$= abc^2 \left[ \frac{n+1}{4n+2} + \frac{n}{4n+2} \cdot \frac{1}{1+x} \Big|_0^{+\infty} \right] = \frac{abc^2}{4n+2}$$

$$3. A = \frac{1}{2} \int_{\partial D} x dy - y dx = \frac{1}{2} \int_a^b \psi(t) d\psi(t) - \psi(t) d\psi(t) = \frac{1}{2} \int_a^b (\psi(t) \psi'(t) - \psi(t) \psi'(t)) dt$$

$$= \frac{1}{2} \int_a^b \begin{vmatrix} \psi(t) & \psi(t) \\ \psi'(t) & \psi'(t) \end{vmatrix} dt$$

$$4. (1) \int_{\mathbb{R}} f(xy) (y dx + x dy) = \int_{\mathbb{R}} f(xy) d(xy) = F(xy) \Big|_{\vec{a}}^{\vec{b}} = 0$$

$$(2) \int_{\mathbb{R}} f(x^2+y^2) (x dx + y dy) = \frac{1}{2} \int_{\mathbb{R}} f(x^2+y^2) d(x^2+y^2) = 0$$

$$(3) \int_{\mathbb{R}} f(x^n+y^n) (x^{n-1} dx + y^{n-1} dy) = \frac{1}{n} \int_{\mathbb{R}} f(x^n+y^n) d(x^n+y^n) = 0$$

### 練習題 18.1

$$1. \int_D d\sigma = \iint_{(x-y)^2 \leq 1} \sqrt{\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} + 1} dx dy = \sqrt{2} \iint_{(x-y)^2 \leq 1} dx dy = \sqrt{2} \cdot \pi$$

$$2. \begin{cases} x = a \cos \theta \\ z = a \sin \theta \end{cases}, y = r \sin \theta, r \in [0, a]$$

$$\int_D d\sigma = \int_0^{2\pi} d\theta \int_0^a \sqrt{\sin^2 \theta \cdot (a^2 + r^2 \cos^2 \theta) - r^2 \cos^2 \theta \sin^2 \theta} dr = \int_0^{2\pi} d\theta \int_0^a \sqrt{a^2 \sin^2 \theta} dr = a^2 \int_0^{2\pi} |\sin \theta| d\theta$$

$$= 2a^2 \int_0^{\pi} \sin \theta d\theta = 4a^2$$

$$3. \begin{cases} x = a \cos \theta \\ y = a \sin \theta \end{cases}, z = t a \cos \theta, t \in [-1, 1]$$

$$\int_D d\sigma = \int_0^{2\pi} d\theta \int_{-1}^1 \sqrt{a^2 \cos^2 \theta \cdot (a^2 + t^2 a^2 \sin^2 \theta) - a^2 \cos^2 \theta \cdot t^2 a^2 \sin^2 \theta} dt = 2a^2 \int_0^{2\pi} |\cos \theta| d\theta = 2a^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos \theta d\theta$$

$$= 4a^2$$

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4. 設  $x = a \cos \theta$ ,  $y = a \sin \theta \cos \varphi$ ,  $z = a \sin \theta \sin \varphi$

其中  $\frac{a^2}{b^2} \sin^2 \theta \cos^2 \varphi \leq \sin^2 \theta \Leftrightarrow \cos^2 \varphi \leq \frac{b^2}{a^2} \Leftrightarrow -\frac{b}{a} \leq \cos \varphi \leq \frac{b}{a}$

$$\begin{aligned} \int_D d\sigma &= 2 \int_0^{2\pi} d\theta \int_{\arccos(\frac{b}{a})}^{\pi - \arccos(\frac{b}{a})} \sqrt{a^2 - a^2 \sin^2 \theta - (a \cos \theta \cos \varphi - (-a \sin \theta \sin \varphi) + a \cos \theta \sin \varphi \cdot a \sin \theta \cos \varphi)^2} d\varphi \\ &= 2a^2 \int_0^{2\pi} d\theta \int_{\arccos(\frac{b}{a})}^{\pi - \arccos(\frac{b}{a})} |\sin \theta| d\varphi = 8a^2 \cdot (\pi - 2\arccos(\frac{b}{a})) = 16a^2 \arcsin(\frac{b}{a}) \end{aligned}$$

5. 設  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = \frac{1}{2} \cdot \frac{r^2}{a} \sin 2\theta$ ,  $r \in [0, a]$

$$\begin{aligned} \int_D d\sigma &= \int_0^a dr \int_0^{2\pi} \sqrt{(1 + \frac{r^2}{a^2} \sin^2 2\theta) \cdot (r^2 + \frac{r^4}{a^2} \cos^2 2\theta) - (\cos \theta \cdot (-r \sin \theta) + \sin \theta \cdot r \cos \theta + \frac{r}{a} \sin 2\theta \cdot \frac{r^2}{a} \cos 2\theta)^2} d\theta \\ &= \int_0^a dr \int_0^{2\pi} \sqrt{r^2 + \frac{r^4}{a^2} \cos^2 2\theta + \frac{r^4}{a^2} \sin^2 2\theta} d\theta = \int_0^a dr \int_0^{2\pi} r \sqrt{1 + \frac{r^2}{a^2}} d\theta \\ &= \pi \int_0^a \sqrt{1 + (\frac{r}{a})^2} dr^2 = \pi a^2 \int_0^1 \sqrt{1+t} dt = \pi a^2 \cdot \frac{2}{3} (1+t)^{\frac{3}{2}} \Big|_0^1 = \frac{2}{3} (2\sqrt{2}-1) \pi a^2 \end{aligned}$$

6. 設  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = \frac{r^2}{2a}$

$r^4 \leq 2a^2 r^2 \cos \theta \sin \theta \Leftrightarrow r^2 \leq a^2 \sin 2\theta$ ,  $\theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$

$$\begin{aligned} \int_D d\sigma &= 2 \int_0^{\frac{\pi}{2}} d\theta \int_0^{a\sqrt{\sin 2\theta}} \sqrt{(1 + \frac{r^2}{a^2})(r^2 + \frac{r^4}{a^2}) - (\cos \theta \cdot (-r \sin \theta) + \sin \theta \cdot (r \cos \theta))^2} dr \\ &= 2 \int_0^{\frac{\pi}{2}} d\theta \int_0^{a\sqrt{\sin 2\theta}} r \sqrt{1 + \frac{r^2}{a^2}} dr = a^2 \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sin 2\theta} \sqrt{1+x} dx = \frac{2a^2}{3} \int_0^{\frac{\pi}{2}} [(1+\sin 2\theta)^{\frac{3}{2}} - 1] d\theta \\ &= \frac{2a^2}{3} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [(1+\cos 2\theta)^{\frac{3}{2}} - 1] d\theta = \frac{2a^2}{3} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (\sqrt{2} \cos^3 \theta - 1) d\theta = \frac{2a^2}{3} \left[ \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sqrt{2}}{2} (\cos^3 \theta + 3 \cos \theta) d\theta + \frac{\pi}{2} \right] \\ &= \frac{2a^2}{3} \left[ \frac{\sqrt{2}}{2} \left( \frac{1}{3} \sin 3\theta + 3 \sin \theta \right) \Big|_{-\frac{\pi}{4}}^{\frac{\pi}{4}} + \frac{\pi}{2} \right] = \frac{a^2}{3} \left( 2\sqrt{2} \left( \frac{1}{3} \times \frac{\sqrt{2}}{2} + 3 \times \frac{\sqrt{2}}{2} \right) + \pi \right) = \frac{a^2}{3} \left( \frac{20}{3} + \pi \right) \end{aligned}$$

$$7. \int_0^{2\pi} d\theta \int_0^a \sqrt{1 \cdot (r^2 + h^2) - 0^2} dr = 2\pi \int_0^a \sqrt{r^2 + h^2} dr$$

$$r = h \tan \beta, \quad \beta \in [0, \arctan(\frac{a}{h})]$$

$$\begin{aligned} 2\pi \int_0^a \sqrt{r^2 + h^2} dr &= 2\pi \int_0^{\arctan(\frac{a}{h})} \frac{h}{\cos \beta} \cdot h (\tan^2 \beta + 1) d\beta = 2\pi \int_0^{\arctan(\frac{a}{h})} \frac{1}{(1 - \sin^2 \beta)^{\frac{3}{2}}} d(\sin \beta) \\ &= 2\pi \int_0^{\frac{a}{\sqrt{a^2 + h^2}}} \frac{1}{(1 - t^2)^{\frac{3}{2}}} dt = \textcircled{1} \end{aligned}$$

$$\text{設 } \frac{1}{(1-t^2)^2} = \frac{A}{1+t} + \frac{B}{1-t} + \frac{C}{(1+t)^2} + \frac{D}{(1-t)^2}$$

$$\Leftrightarrow 1 = A(1-t)^2(1+t) + B(1+t)^2(1-t) + C(1-t)^2 + D(1+t)^2$$

$$C = \frac{1}{4}, D = \frac{1}{4}, A = B, A+B = \frac{1}{2}$$

$$\Leftrightarrow A=B=C=D=\frac{1}{4}$$

$$\textcircled{1} = 2\pi \left[ \frac{\pi}{2} \left( \ln \left| \frac{t+1}{t-1} \right| - \frac{1}{t+1} - \frac{1}{t-1} \right) \right]_0^{\frac{a}{\sqrt{a^2+h^2}}} = \frac{\pi}{2} \left( \ln \frac{\sqrt{a^2+h^2} + a}{\sqrt{a^2+h^2} - a} - \frac{\sqrt{a^2+h^2}}{a + \sqrt{a^2+h^2}} - \frac{\sqrt{a^2+h^2}}{a - \sqrt{a^2+h^2}} \right)$$

## 練習題 18.2

$$(2) \text{ 記 } x = \sqrt{z} \cos \theta, \quad y = \sqrt{z} \sin \theta$$

$$\begin{aligned} \int_{\frac{z}{2}} |xy z| d\sigma &= \int_0^{2\pi} d\theta \int_0^1 z^2 |\cos \theta \sin \theta| \sqrt{\left[ \left( \frac{\cos \theta}{2\sqrt{z}} \right)^2 + \left( \frac{\sin \theta}{2\sqrt{z}} \right)^2 + 1 \right] \left[ (\sqrt{z})^2 \right]} dz \\ &= \int_0^{2\pi} d\theta \int_0^1 \frac{z^2}{2} |\sin 2\theta| \sqrt{z + \frac{1}{4}} dz = \left( \int_0^{2\pi} |\sin 2\theta| d\theta \right) \cdot \frac{1}{2} \left( \int_0^1 z^2 \sqrt{z + \frac{1}{4}} dz \right) \end{aligned}$$

$$= 4 \int_0^{\frac{\pi}{2}} \sin 2\theta d\theta \cdot \frac{1}{2} \int_{\frac{1}{4}}^{\frac{5}{4}} \left( t - \frac{1}{4} \right)^2 \sqrt{t} dt$$

$$= 2 \int_{\frac{1}{4}}^{\frac{5}{4}} \left( t^{\frac{5}{2}} - \frac{1}{2} t^{\frac{3}{2}} + \frac{1}{16} t^{\frac{1}{2}} \right) dt = 2 \left( \frac{2}{7} t^{\frac{7}{2}} - \frac{1}{5} t^{\frac{5}{2}} + \frac{1}{24} t^{\frac{3}{2}} \right) \Big|_{\frac{1}{4}}^{\frac{5}{4}}$$

$$= 2 \times \left[ \frac{2}{7} \times \left( \frac{5^{\frac{7}{2}}}{2^{\frac{7}{2}}} - \frac{1}{2^{\frac{7}{2}}} \right) - \frac{1}{5} \times \left( \frac{5^{\frac{5}{2}}}{2^{\frac{5}{2}}} - \frac{1}{2^{\frac{5}{2}}} \right) + \frac{1}{24} \times \left( \frac{5^{\frac{3}{2}}}{2^{\frac{3}{2}}} - \frac{1}{2^{\frac{3}{2}}} \right) \right]$$

$$= \frac{125\sqrt{5}}{224} - \frac{1}{224} - \frac{5\sqrt{5}}{32} + \frac{1}{160} + \frac{5\sqrt{5}}{192} - \frac{1}{192}$$

$$= \frac{45\sqrt{5}}{112} + \frac{5\sqrt{5}}{192} + \frac{1}{32} \times \left( \frac{1}{5} - \frac{1}{6} - \frac{1}{7} \right)$$

$$= \frac{5\sqrt{5}}{16} \left( \frac{9}{7} - \frac{1}{12} \right) + \frac{1}{32} \times \left( \frac{1}{30} - \frac{1}{7} \right) = \frac{505\sqrt{5}}{1344} - \frac{23}{6720}$$



姓名：

( 號)

成績

年級：

科目：

(3) 設  $x = z \cos \theta$ ,  $y = z \sin \theta$

$$z^2 \leq 2z \cos \theta \Leftrightarrow 0 \leq z \leq 2 \cos \theta$$

$$\begin{aligned} \int_{\Sigma} (xy + yz + zx) d\sigma &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{2 \cos \theta} z^2 (\cos \theta \sin \theta + \sin \theta + \cos \theta) \sqrt{(1+1)(z^2)} dz \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{2} \cdot \frac{1}{4} (2 \cos \theta)^4 (\cos \theta \sin \theta + \cos \theta + \sin \theta) d\theta \\ &= 4\sqrt{2} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^5 \theta d\theta = 8\sqrt{2} \int_0^{\frac{\pi}{2}} (1 - \sin^4 \theta)^2 d(\sin \theta) \\ &= 8\sqrt{2} \int_0^1 (1 - x^4)^2 dx = 8\sqrt{2} \int_0^1 (x^4 - 2x^2 + 1) dx \\ &= 8\sqrt{2} \times \left( \frac{1}{5} - \frac{2}{3} + 1 \right) = \frac{64\sqrt{2}}{15} \end{aligned}$$

### 問題 18.2

1. 注意到  $ax + by + cz = [a \ b \ c] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \vec{x} \cdot \vec{c}$ , 且  $\| [a \ b \ c] \| = \sqrt{a^2 + b^2 + c^2}$ ,  $\| [x \ y \ z] \| = \sqrt{x^2 + y^2 + z^2} = 1$

考慮到  $abc \neq 0$ ,

記  $\vec{e} = \frac{1}{\sqrt{a^2 + b^2 + c^2}} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ , 必存在唯一一個正交矩陣  $Q$ , 使得  $Q\vec{e} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$\text{則 } f(ax + by + cz) = f(\sqrt{a^2 + b^2 + c^2} \vec{e}^T \vec{x}) = f(\sqrt{a^2 + b^2 + c^2} \vec{e}^T Q^T Q \vec{x})$$

$$\text{記 } Q\vec{x} = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}, f(\sqrt{a^2 + b^2 + c^2} \cdot [1 \ 0 \ 0] \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}) = f(\sqrt{a^2 + b^2 + c^2} x')$$

$$\begin{aligned} E(\vec{x}) &= \left( \frac{\partial x}{\partial u} \right)^2 + \left( \frac{\partial y}{\partial u} \right)^2 + \left( \frac{\partial z}{\partial u} \right)^2 = \left[ \frac{\partial}{\partial u} [x \ y \ z] \right] \left[ \frac{\partial}{\partial u} [x \ y \ z] \right]^T \\ &= \left[ \frac{\partial}{\partial u} [x' \ y' \ z'] \right] Q^T Q \left[ \frac{\partial}{\partial u} [x' \ y' \ z'] \right]^T \\ &= \left[ \frac{\partial}{\partial u} [x' \ y' \ z'] \right] \left[ \frac{\partial}{\partial u} [x' \ y' \ z'] \right]^T = E(Q\vec{x}) \end{aligned}$$

$$\begin{aligned} \text{同理有 } E(\vec{x}) &= E(Q\vec{x}), F(\vec{x}) = \left[ \frac{\partial}{\partial u} [x \ y \ z] \right] \left[ \frac{\partial}{\partial v} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right] = \left[ \frac{\partial}{\partial u} [x' \ y' \ z'] \right] Q Q^T \left[ \frac{\partial}{\partial v} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \right] \\ &= F(Q\vec{x}) \end{aligned}$$